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Inventor(s)

Wilson; Stephen S. et al.

MULTI-LAYER WINDSHIELD FILM HAVING PROGRESSIVE THICKNESS LAYERS

Abstract

A protective barrier is affixable to a curved substrate such as a vehicle windshield. The protective barrier may include a stack of three or more lenses. Each of the three or more lenses may include a thermoplastic film (preferably a polyethylene terephthalate (PET) film) and an adhesive layer on a first side of the thermoplastic film. The three or more lenses may have respective thicknesses that define a monotonically decreasing function in a stacking direction of the stack and that include at least three different thicknesses. The protective barrier may be placed on the curved substrate with the adhesive layer of an innermost lens of the stack in contact with the curved substrate, and heat and pressure may be applied to conform the stack to the shape of the curved substrate. Thereafter, an outermost lens of the stack may be peeled off to reveal the next lens of the stack.

Inventors: Wilson; Stephen S. (Las Vegas, NV), Wilson; Bart E. (Las Vegas, NV), Fremont; Christophe (Las Vegas, NV)

Applicant: RO Technologies, LLC (Las Vegas, NV)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. application Ser. No. 18/943,256, filed Nov. 11, 2024, which is a continuation of U.S. application Ser. No. 17/937,371, filed Sep. 30, 2022 and issued as U.S. Pat. No. 12,162,330 on Dec. 10, 2024, which relates to and claims the benefit of U.S. Provisional Application No. 63/267,686, filed Feb. 8, 2022 and entitled “MULTI-LAYER WINDSHIELD FILM HAVING PROGRESSIVE THICKNESS LAYERS,” the entire contents of which are expressly incorporated by reference.

STATEMENT RE: FEDERALLY SPONSORED RESEARCH/DEVELOPMENT

[0002] Not Applicable

BACKGROUND

[0003] For participants in motorsports such as stock car racing and off-road racing, it is important to maintain visibility of the course as mud, bugs, and other debris accumulate on the windshield of the vehicle and as the windshield becomes pitted by hard debris such as rocks. The same is true, though on a more gradual scale, for ordinary street vehicles. In addition, auto racers and ordinary street vehicle owners alike have a need to protect the underlying windshield from pitting or cracking in view of the high expense and inconvenience of replacing it. Indeed, this aspect of the problem is even more pronounced in the case of street vehicles due to the much greater number of street vehicles in existence and the higher cost of glass windshields in comparison to polycarbonate windshields as are typically used in racecars. To meet these needs, tearoff films are commonly employed, either individually or in laminated stacks, which may be applied to any type of vehicle windshield. As debris accumulates on the outermost tearoff film, or as the outermost film becomes pitted and otherwise damaged, the driver simply tears it off to reveal the next pristine film underneath. Depending on the number of films in the stack, the process may be repeated several times before replacing the stack.

[0004] It has been found by the present inventors that much of the protective benefit of a laminated stack of tearoff films comes from its total thickness as it serves to cushion the impact of hard debris. Unfortunately, this potential benefit is at odds with its functionality as a laminated stack since each removed film dramatically reduces the thickness of the stack, resulting in a reduced resistance to impact and a significantly increased likelihood that the underlying windshield will become damaged by debris while the stack is in use. Moreover, there is a performance tradeoff between the impact cushioning effect and the increased distortion associated with the use of relatively thicker films.

BRIEF SUMMARY

[0005] The present disclosure contemplates various systems and methods for overcoming the above

drawbacks accompanying the related art. One aspect of the embodiments of the present disclosure is a protective barrier affixable to a curved substrate. The protective barrier may comprise a stack of three or more lenses. Each of the three or more lenses may include a thermoplastic film (preferably a polyethylene terephthalate (PET) film or a thermoplastic polyurethane (TPU) film) and an adhesive layer on a first side of the thermoplastic film. The three or more lenses may have respective thicknesses that define a monotonically decreasing function in a stacking direction of the stack and that include at least three different thicknesses.

[0006] The thermoplastic films of the three or more lenses may have respective thicknesses that define a monotonically decreasing function in a stacking direction of the stack and that include at least three different thicknesses. The thermoplastic film of an innermost lens of the stack may be at least twice as thick as the thermoplastic film of an outermost lens of the stack, preferably at least three times as thick.

[0007] The thermoplastic film of an innermost lens of the stack may have a thickness of five mil or greater. The thickness of the thermoplastic film of the innermost lens of the stack may be seven mil or less, preferably six mil or less.

[0008] The thermoplastic film of an outermost lens of the stack may have a thickness of three mil or less. The thickness of the thermoplastic film of the outermost lens of the stack may be two mil or greater.

[0009] The adhesive layers of the three or more lenses may have respective thicknesses that define a monotonically decreasing function in a stacking direction of the stack and that include at least three different thicknesses. The adhesive layer of an innermost lens of the stack may be at least twice as thick as the adhesive layer of an outermost lens of the stack. The adhesive layer of the innermost lens of the stack may be at least three times as thick as the adhesive layer of the outermost lens of the stack.

[0010] The thermoplastic films of the three or more lenses may have respective thicknesses that include at least two different thicknesses.

[0011] The adhesive layers of the three or more lenses may have respective thicknesses that include at least two different thicknesses.

[0012] The three or more lenses may number at least four. At least two of the lenses may have the same thickness.

[0013] A total thickness of the stack may be between ten mil and thirty mil. The total thickness of the stack may be between fifteen mil and twenty-five mil.

[0014] Another aspect of the embodiments of the present disclosure is a system. The system may comprise a vehicle windshield and a protective barrier affixed to the vehicle windshield. The protective barrier may comprise a stack of three or more lenses. Each of the three or more lenses may include a thermoplastic film (preferably a polyethylene terephthalate (PET) film or a thermoplastic polyurethane (TPU) film) and an adhesive layer on a first side of the thermoplastic film. The three or more lenses may have respective thicknesses that define a monotonically decreasing function in a stacking direction of the stack and that include at least three different thicknesses.

[0015] The vehicle windshield may be a windshield of a street vehicle.

[0016] Another aspect of the embodiments of the present disclosure is a method. The method may comprise stacking three or more lenses. Each of the three or more lenses may include a thermoplastic film (preferably a polyethylene terephthalate (PET) film or a thermoplastic polyurethane (TPU) film) and an adhesive layer on a first side of the thermoplastic film. The three or more lenses may have respective thicknesses that define a monotonically decreasing function in a stacking direction of the stack and that include at least three different thicknesses. The method may further comprise placing the stack of three or more lenses on a curved substrate with the adhesive layer of an innermost lens of the stack in contact with the curved substrate and applying heat and pressure to conform the stack of three or more lenses to the shape of the curved substrate.

[0017] The method may comprise peeling off an outermost lens of the stack of three or more lenses after said applying heat and pressure.

[0018] Another aspect of the embodiments of the present disclosure is a protective barrier affixable to a curved substrate. The protective barrier may comprise a stack of two or more lenses. Each of the two or more lenses may include a thermoplastic film (preferably a polyethylene terephthalate (PET) film or a thermoplastic polyurethane (TPU) film) and an adhesive layer on a first side of the thermoplastic film. The adhesive layers of the two or more lenses may have respective thicknesses that define a monotonically decreasing function in a stacking direction of the stack and that include at least two different thicknesses.

[0019] Another aspect of the embodiments of the present disclosure is a protective barrier affixable to a curved substrate, the protective barrier comprising a stack of one or more lenses, each of the one or more lenses including a thermoplastic polyurethane (TPU) film and an adhesive layer on a first side of the TPU film.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0020] These and other features and advantages of the various embodiments disclosed herein will be better understood with respect to the following description and drawings, in which like numbers refer to like parts throughout, and in which:

[0021] FIG. 1 is a cross-sectional view of a protective barrier according to an embodiment of the present disclosure;

[0022] FIG. 2 is an exploded view of the protective barrier;

[0023] FIG. 3 is a cross-sectional view of another protective barrier according to an embodiment of the present disclosure; and

[0024] FIG. 4 is an example operational flow according to an embodiment of the present disclosure.

DETAILED DESCRIPTION

[0025] The present disclosure encompasses various embodiments of a protective barrier affixable to a curved substrate such as a vehicle windshield and methods of manufacture, installation, and use thereof. The detailed description set forth below in connection with the appended drawings is intended as a description of several currently contemplated embodiments and is not intended to represent the only form in which the disclosed invention may be developed or utilized. The description sets forth the functions and features in connection with the illustrated embodiments. It is to be understood, however, that the same or equivalent functions may be accomplished by different embodiments that are also intended to be encompassed within the scope of the present disclosure. It is further understood that relational terms such as first and second and the like are used solely to distinguish one from another entity without necessarily requiring or implying any actual such relationship in order between such entities.

[0026] FIG. 1 is a cross-sectional view of a protective barrier **100** according to an embodiment of the present disclosure. The protective barrier **100** may be affixed to a substrate **10** such as a windshield of an automobile and may comprise a stack of three or more lenses **110** (or in some cases two or more lenses **110**) such as the lenses **110a**, **110b**, **110c** shown in FIG. 1. Each of the lenses **110** may include a thermoplastic film (preferably a polyethylene terephthalate (PET) film) **112** and an adhesive layer **114** on a first side thereof for bonding the lenses **110** together and to the substrate **10**. Unlike conventional stacks of tearoff films, the three or more lenses **110** may have respective thicknesses that define a monotonically decreasing function (i.e., one that either decreases or stays the same but does not increase) in a stacking direction of the stack (i.e., from a lens **110** nearest the substrate **10** toward a lens **110** farthest from the substrate **10**). The respective

thicknesses of the three or more lenses **110** may further include at least three different thicknesses arranged accordingly, with each change in thickness thus being a decrease, resulting in thinner lenses **110** being arranged later in the stacking direction (though the stack may include adjacent lenses **110** whose respective thicknesses are the same). As shown by way of example in FIG. **1**, the three illustrated lenses **110a**, **110b**, **110c** may have respective thicknesses D_a , D_b , D_c that are all different and which decrease from a first (innermost) lens **110a** closest to the substrate **10** to a last (outermost) lens **110c** farthest from the substrate **10**, i.e., $D_{sub.a} > D_{sub.b} > D_{sub.c}$. Owing to the described construction, as compared to conventional stacks where all layers have the same thickness, the protective barrier **100** may maintain a greater total remaining thickness as each lens **110** is removed. As a result, the peeling away of each lens **110** during the life of the protective barrier **100** does not degrade the cushioning effect of the stack as much, affording better protection of the substrate **10** against harmful impacts. At the same time, the novel construction of the protective barrier **100** makes it possible to achieve a desired performance balance by combining the cushioning effect of relatively thicker films with the reduced distortion of relatively thinner films.

[0027] FIG. **2** is an exploded view of the protective barrier **100** shown in FIG. **1**. In general, the different thicknesses D_a , D_b , D_c may be achieved by adjusting the thickness of any sub-layer of each lens **110**. Thus, in the illustrated example, where each lens **110** has a thermoplastic film **112** and an adhesive layer **114**, the different thicknesses D_a , D_b , D_c may be the result of varying the thickness(es) of one or both of these two layers **112**, **114** in each lens **110**. For the sake of example, FIGS. **1** and **2** show one such possibility in which both the thickness of the thermoplastic film **112** and the thickness of the adhesive layer **114** decrease in the stacking direction. In particular, in the illustrated example, the respective thickness of the thermoplastic films **112**, identified as $d_{sub.1a}$, $d_{sub.1b}$, and $d_{sub.1c}$ in FIG. **2**, have the relationship $d_{sub.1a} > d_{sub.1b} > d_{sub.1c}$, and the respective thicknesses of the adhesive layers, identified as $d_{sub.2a}$, $d_{sub.2b}$, and $d_{sub.2c}$ in FIG. **2**, likewise have the relationship $d_{sub.2a} > d_{sub.2b} > d_{sub.2c}$. However, either of these relationships may, on its own, produce the relationship $D_{sub.a} > D_{sub.b} > D_{sub.c}$, with the thicknesses of the other layer (i.e., the thicknesses of the thermoplastic films **112** or the thicknesses of the adhesive layers **114**) remaining constant or having a different relationship, for example.

[0028] More generally, and accounting for embodiments in which there may be adjacent lenses **110** having the same thickness, the thermoplastic films **112** of the three or more lenses **110** may have respective thicknesses that define a monotonically decreasing function in the stacking direction. Additionally, or alternatively, the adhesive layers **114** of the three or more lenses **110** may have respective thicknesses that define a monotonically decreasing function in the stacking direction. In addition, either the respective thicknesses of the thermoplastic films **112** or the respective thicknesses of the adhesive layers **114**, or both, may include at least two different thicknesses or, in some cases, three or more thicknesses. For example, one practical implementation may be to produce thermoplastic films **112** of three or more different thicknesses and to stack them so that the thermoplastic films **112** have monotonically decreasing thickness in the stacking direction, possibly keeping the thicknesses of the adhesive layers **114** constant. In such a stack, the thermoplastic film **112** of an innermost lens **110** of the stack may be at least twice as thick as the thermoplastic film **112** of an outermost lens **110** of the stack, for example, preferably at least three times as thick. Another practical implementation may be to produce thermoplastic films **112** having the same thickness but to apply differing amounts of adhesive thereto so that the adhesive layers **114** have monotonically decreasing thickness in the stacking direction. In such a stack, the adhesive layer **114** of an innermost lens **110** of the stack may be at least twice as thick as the adhesive layer **114** of an outermost lens **110** of the stack, for example, preferably at least three times as thick. Between these two extremes, the thicknesses of the thermoplastic films **112** or the thicknesses of the adhesive layers **114** may transition gradually from the thickest thermoplastic film **112** or thickest adhesive layer **114** of the innermost lens **110** to the thinnest thermoplastic film **112** or thinnest adhesive layer **114** of the outermost lens **110**. Preferably, and as described in more detail below, the

thicknesses of the thermoplastic films **112** or adhesive layers **114** may transition not perfectly evenly but in discrete stages in order to maximize the remaining total thickness of the stack as much as possible as each lens **110** is removed.

[0029] FIG. 3 is a cross-sectional view of another protective barrier **300** according to an embodiment of the present disclosure. Like the protective barrier **100**, the protective barrier **300** may be affixed to a substrate **10** such as a windshield of an automobile and may comprise a stack of three or more lenses **110** (or in some cases two or more lenses **110**), in this case the lenses **110a**, **110b**, and **110c** and an additional lens **110c**. Each of the lenses **110** may be the same as those described in relation to FIG. 1 and may include a thermoplastic film (preferably a polyethylene terephthalate (PET) film) **112** and an adhesive layer **114** on a first side thereof for bonding the lenses **110** together and to the substrate **10**. The example of the protective barrier **300** of FIG. 3 is provided in order to illustrate that the monotonically decreasing function described above may allow for adjacent lenses **110** whose respective thicknesses are the same. In this regard, FIG. 3 illustrates the specific case where the thinnest lens **110c** (having thickness D_e) is repeated. More generally, any of the lenses **110** may be repeated any number of times within the meaning of the monotonically decreasing function contemplated herein. For example, there may be two lenses **110a**, two lenses **110b**, and two lenses **110c**, or there may be three lenses **110a**, one lens **110b**, and three lenses **110c**. Thus, for example, the three or more lenses **110** contemplated may number four or more, with two (or more) of the lenses having the same thickness. It should also be noted that there is no limitation on the number of differently fabricated lenses **110** having different thicknesses. That is, there may also be one or more lenses **110d** with thickness $D_{sub.d}$ (where $D_{sub.d} < D_{sub.c}$), one or more lenses **110e** with thickness $D_{sub.e}$ (where $D_{sub.e} < D_{sub.d}$), etc. Typically, however, the total number of lenses **110** may be five or less so as not to introduce too much optical distortion to the protective barrier **100**.

[0030] In general, the thicknesses of the lenses **110** of the protective barriers **100**, **300** contemplated herein may transition evenly from the thickest (innermost) lens **110** to the thinnest (outermost) lens **110** or may, preferably, transition in discrete stages in order to maximize the remaining total thickness of the stack as much as possible as each lens **110** is removed. As a numeric example, the following Table 1 compares the remaining stack thickness in mil as each lens **110** is removed in three hypothetical protective barriers. In each case, the starting total thickness of the stack is 20.00 mil and there are four lenses **110**. Though the specific constraints on each lens's thickness may depend on the quality or strength of PET used (e.g., higher quality PET may allow for thinner thermoplastic films **112**) and on the particular application (e.g., thicker thermoplastic films **112** may be usable when a higher degree of optical distortion is acceptable), it is assumed in each case that the thinnest usable thermoplastic film **112** is 2.00 mil and the thickest usable thermoplastic film **112** is 7.00 mil. (However, it is noted that stronger PET may make 1.00 mil thermoplastic films **112** possible in some cases.) For purposes of this example, it is also assumed that the thickness of the adhesive layer **114** is kept constant at 0.50 for each lens **110**, with the total thickness of each lens **110** thus ranging from 2.50 mil to 7.50 mil.

TABLE-US-00001

TABLE 1	Constant	Progressive	Progressive	Lens #	Thickness	Thickness	1
Thickness 2	4 (outermost)	5.00 (20.00 left)	2.50 (20.00 left)	2.50 (20.00 left)	3	5.00 (15.00 left)	
4.17 (17.50 left)	3.00 (17.50 left)	2	5.00 (10.00 left)	5.83 (13.33 left)	7.00 (14.50 left)	1 (innermost)	
5.00 (5.00 left)	7.50 (7.50 left)	7.50 (7.50 left)					

[0031] In Table 1, the protective barrier labeled “Constant Thickness” has four equally thick lenses **110** of 5.0 mil each, while the two protective barriers labeled “Progressive Thickness 1” and “Progressive Thickness 2” exhibit monotonically decreasing thickness from the innermost lens to the outermost lens including at least three thicknesses as described herein. As can be seen from a comparison between the different hypothetical protective barriers in Table 1, the total remaining thickness in the “Progressive Thickness 1” and “Progressive Thickness 2” examples is significantly greater than that in the “Constant Thickness” example as each lens **110** is removed, such that the

remaining stack provides a superior cushioning effect to withstand impacts. For example, after the outermost lens (lens number 4) is removed, the “Constant Thickness” stack has gone from a total thickness of 20.00 mil down to a total thickness of 15.00 mil, while the two “Progressive Thickness” examples have only dropped to a total thickness of 17.50 mil. The “Progressive Thickness” examples continue to preserve more total thickness as the lenses **110** are removed. The monotonic decrease in thickness of “Progressive Thickness 1” represents an even transition in which the change in thickness is constant from one lens **110** to the next, while the monotonic decrease in thickness of “Progressive Thickness 2” represents an uneven transition marked by discrete stages, such that there is a stage of relatively thin lenses **110** on the outermost part of the stack (2.50 mil and 3.00 mil) and a stage of relatively thick lenses **110** on the innermost part of the stack (7.00 mil and 7.50 mil). By arranging the thicknesses of the lenses **110** in this way, the remaining thickness as each lens **110** is removed can be further optimized. As can be seen, for example, “Progressive Thickness 2” retains 14.50 mil total thickness after the second lens **110** is removed, while “Progressive Thickness 1” retains only 13.33 mil total thickness (though this is still far superior to the 10.00 mil total thickness retained by the “Constant Thickness” example). In this way, an uneven transition between the thickest and thinnest lenses **110** may result in a greater total thickness remaining after each lens **110** is removed, further optimizing the protective barrier **100** for protection against impacts during use.

[0032] In the above examples described in relation to Table 1, the thicknesses of the adhesive layers **114** are assumed to be kept constant at 0.50 mil, with the thicknesses of the thermoplastic films **112** being 4.50 mil in the “Constant Thickness” example and transitioning from 7.00 mil to 2.00 mil in the two examples of progressive thickness (“Progressive Thickness 1” and “Progressive Thickness 2”). In general, when implementing the disclosed monotonic decrease in thickness, it is contemplated that the thermoplastic film **112** of the innermost lens **110** of the stack, as well as the thermoplastic films **112** of those lenses **110** nearest it, may preferably be relatively thick (e.g., five mil or greater) so that the greatest portion of the total thickness of the protective layer **100** is concentrated at the innermost part of the stack. At the same time, the thickness of this innermost thermoplastic film **112** preferably does not exceed seven mil (and more preferably does not exceed six mil) in order to keep the degree of optical distortion within a range that is suitable for a windshield use case. As for the thermoplastic film **112** of the outermost lens **110** of the stack and those lenses **110** nearest it, the thickness is preferably three mil or less so that the first lenses **110** that are removed from the stack do not significantly reduce the total thickness of the stack. At the same time, the thickness of the outermost thermoplastic film **112** (i.e., the minimum film thickness) is typically at least two mil or greater so that it is strong enough to withstand the forces associated with the user peeling lenses **110** from the stack during use. It is noted, however, that since the outermost thermoplastic film **112** does not need to undergo the pulling force of any previous (i.e., higher) adhesive layer, the thermoplastic film **112** of the outermost lens **110** may in some cases be significantly thinner than the thermoplastic film **112** of the lens **110** underneath it.

[0033] As noted above, the different thicknesses D.sub.a, D.sub.b, D.sub.c may be achieved by adjusting the thickness of any sub-layer of each lens **110**, including the adhesive layers **114**. In this regard, the thickness of the adhesive layer **114** may typically be varied between a minimum thickness that is 0.050 mil or greater and a maximum thickness that is 2.00 mil or less, for example.

[0034] FIG. 4 is an example operational flow according to an embodiment of the present disclosure. The operational flow of FIG. 4 may serve as an example method of manufacturing, installing, and using the protective barrier **100**, **300** including the stack of lenses **110** shown and described in relation to FIGS. 1-3. The operational flow may begin with providing the thermoplastic film **112** to be used in each of three or more lenses **110** (step **410**). The thermoplastic film **112** of each of the lenses **110** may be a biaxially-oriented polyethylene terephthalate (BoPET) and may be selected for particular MTF data or may be fabricated while actively monitoring the MTF data in a continuous or batch-to-batch process as described in any of U.S. Patent Application

Pub. No. 2021/0162645, entitled “Method and Apparatus for Reducing Non-Normal Incidence Distortion in Glazing Films,” U.S. Patent Application Pub. No. 2021/0283994, entitled “Protective Barrier for Safety Glazing,” or commonly owned U.S. patent application Ser. No. 17/505,433, entitled “Method and Apparatus for Reducing Non-Normal Incidence Distortion in Glazing Films,” the entire contents of each of which is incorporated by reference herein. In this regard, providing the thermoplastic film **112** may include, for example, melting a resin, extruding the melted resin through a die to produce a polymer film, and cooling the polymer film. Each thermoplastic film **112** may be coated on one side thereof with an adhesive layer **114** (step **420**), which is preferably wet deposited but may be applied according to any appropriate methods including spin coating, dip coating, or vacuum deposition. The adhesive layer **114** may be a wet mount adhesive as disclosed, for example, in U.S. Pat. No. 9,128,545, entitled “Touch Screen Shield” or a dry mount adhesive as disclosed, for example, in U.S. Pat. No. 9,295,297, entitled “Adhesive Mountable Stack of Removable Layers,” the entire contents of each of which is incorporated by reference herein. The adhesive **114** may be an acrylic or silicon adhesive such as an acrylic pressure sensitive adhesive (PSA) or a silicon PSA.

[0035] With the thermoplastic films **112** having been coated with the adhesive layers **114**, the operational flow of FIG. **4** may continue with stacking three or more of the resulting lenses **110** to produce a progressive thickness stack (step **430**). For example, as described above, either the thickness of the thermoplastic films **112**, the thickness of the adhesive layers **114**, or both may be varied according to a monotonically decreasing function from an innermost lens **110** to an outermost lens **110**, such that the thickness of each lens **110** is either less than or equal to that of the previous lens **110** in the stacking direction. In addition to the thermoplastic film **112** and adhesive layer **114**, it is contemplated that each lens **110** may further include other layers, such as a hard coat as described in the above-mentioned '994 publication, a UV blocking layer, or a thermochromic film as described in U.S. Patent Application Pub. No. 2021/0070017, entitled “Nano Particle Solar Control Film,” the entire contents of which is incorporated by reference herein. The thicknesses of any such additional layers may likewise be controlled in order to maintain the monotonically decreasing thickness described herein. The total thickness of the stack of lenses **110** may preferably be between ten mil and thirty mil, more preferably between fifteen mil and twenty-five mil (e.g., twenty mil as in the above examples of Table 1). In order to keep optical distortion as low as possible, the refractive indices of the thermoplastic films **112**, the adhesive layers **114**, and any other layers within each lens **110** and between each of the plurality of lenses **110** may be matched to within 0.2 as described in the above-mentioned '297 patent.

[0036] Once the protective barrier **100** comprising the stack of lenses **110** has been assembled, the operational flow may continue with installing the protective barrier **100** on a curved substrate **10** such as the windshield of a car or other vehicle. Continuing to refer to the operational flow of FIG. **4**, the protective barrier **100**, including the stack of lenses **110**, may be placed on the windshield or other curved substrate **10** (step **440**), with the adhesive layer **114** of the lowermost lens **110a** (see FIGS. **1-3**) in contact with the curved substrate **10**. For easier installation, the protective barrier **100** may be rough cut (e.g. using an electric film cutter) so as not to extend too far outside the windshield **10**. The operational flow may continue with applying heat and pressure to conform the stack of three or more lenses **110** to the shape of the curved substrate **10** (step **450**). In particular, the applying of heat and pressure may be performed at least in part prior to the adhesive layer **114** of each of the three or more lenses **110** being fully cured, for example, prior to the adhesive layer **114** exceeding a peel strength of 25 grams per inch determined as a constant load per unit width needed for peeling. The protective barrier **100** may be completely conformed to the shape of the curved substrate **10** prior to the adhesive layers **114** being fully cured. In some cases, the protective barrier **100** may be applied to the substrate **10** using a sacrificial layer serving as a female mold cavity to sandwich the stack of lenses **110** between the sacrificial layer and the substrate **10** as described in U.S. Patent Application Pub. No. 2020/0247102, entitled “Thermoform Windshield

Stack with Integrated Formable Mold,” the entire contents of which is incorporated by reference herein. After allowing the protective barrier **100** to cool down, the installation may conclude with performing a final trim to fit the windshield or other substrate **10**.

[0037] As explained above, it is contemplated that a protective barrier **100** having more than one lens **110** may allow for the outermost lens **110** to be peeled off and removed to reveal the unused surface of the lens **110** beneath. In this respect, the operational flow of FIG. **4** may continue during the life of the protective barrier **100** that has been installed on a vehicle windshield or other substrate **10**. When the outermost lens **110** becomes unacceptably degraded over time (e.g. after six months, after a year, after scratching from wiper blades begins to occur, etc.), it may be peeled off to reveal the next lens **110** underneath (step **460**). The timing of peeling off the outermost lens **110** may depend on the particular climate where the protective barrier **100** is used, with some climates entailing more exposure to sun and others requiring more frequent use of wiper blades, for example. Owing to the progressive thickness of the stack of lenses **110**, the total thickness of the protective barrier **100** may remain significantly greater than a conventional stack of tearoff lenses after the removal of the outermost lens **110**, resulting in a longer lasting cushioning effect against impacts during the life of the protective barrier **100**.

[0038] In the above disclosure, it is noted that PET (e.g., BoPET) may be a preferred thermoplastic film **112**. However, the material of the thermoplastic film **112** is not necessarily limited in this respect. For example, the thermoplastic film **112** of any or all of the lenses **110** may instead be a thermoplastic polyurethane (TPU) film, such as an optically clear TPU film. The TPU film may be manufactured to have a visible light transmittance (VLT) of greater than 85% (e.g., 90-95%) while having a low haze (e.g., 1% or lower). The TPU film may be greater than 2 mils, for example. Advantageously, the use of TPU for the thermoplastic film **112** of the protective barrier **100** may allow for a quicker installation process, requiring less expertise on the part of the installer, as compared to when PET is used. This is due to TPU's ease of applicability on 2D or 3D curved surfaces, without there being the need to apply heat to thermoform the film or the stack, or with only the need to apply very little heat to thermoform the TPU in seconds when the film is to be installed. In particular, TPU is more stretchable than PET and also requires less heat to become increasingly more stretchable. Common issues, such as fingering (i.e., imperfect application resulting in finger-shaped bubbles underneath the film), that occur when installing a PET-based stack require a specific thermoforming technique to avoid and/or correct. Correcting these imperfections itself is a lengthy process, resulting in longer installation time by a professional. In contrast, TPU with strong elasticity and elongation properties doesn't require as much technique and in some cases may not require any professional technique.

[0039] As a result of the above advantages, the use of TPU for the thermoplastic film(s) **112** may allow for a significantly lower installation cost than that of PET-based thermoplastic film(s) **112**, resulting in a highly affordable product. Such an affordable product is especially important as a protective barrier for the underlying windshield as windshields become increasingly costly due to Advanced Driver Assistance Systems (ADAS) technology, the increased windshield curvature of new car designs, and the larger windshield size of new car designs. It is also contemplated that TPU-based films **112**, with or without a protective hard coat, may last longer than PET-based films **112**. As such, the use of TPU in the disclosed embodiments (as well as for protective films in general, even single-layer stacks as described below) may result in a lower cost film, a significantly shorter installation time (e.g., one-fifth the installation time), a longer lifetime, and a lower cost per windshield per year (e.g., one-tenth of the cost in the case of PET-based protective films) as the underlying windshields are protected from breaking, cracking, or pitting.

[0040] Furthermore, when a windshield is replaced in the U.S., even if it can be recycled by being sent to a third-party recycling company, over 95% of the time the old windshield is not recycled. The vast majority of the estimated 15 million windshields damaged are being dumped in landfills each year in the U.S. alone, resulting in massive pollution. See Wolfel B, Secfried A, Allen V,

Kaschta J, Holmes C, Schubert D W. Recycling and Reprocessing of Thermoplastic Polyurethane Materials towards Nonwoven Processing. *Polymers*. 2020; 12(9):1917.
<https://doi.org/10.3390/polym12091917>. By using a stack of one or more protective TPU films, a significant portion of this waste can be avoided. Meanwhile, the TPU itself can be easily recycled and is a 100% recyclable and biodegradable material. It is an advanced material much more environmentally friendly than alternatives such as PVC since TPU is recyclable and biodegradable in three to five years, in comparison to glass windshield material that may take hundreds of thousands of years to decompose in the landfill assuming it decomposes at all.

[0041] Advantageously, it is contemplated that TPU-based films **112** may replace PET-based films **112** in the disclosed protective barrier **100** as described above. However, the above disclosure regarding the advantages of using TPU-based films **112** is not intended to be limited only to use in the protective barrier **100** described herein. TPU-based films **112** may, for example, be used in other multi-layer stacks of lenses **110** or in single-layer stacks of one lens **110**. A contemplated single-layer stack of one lens **110** may include, for example, the TPU film **112** and the adhesive layer **114** for bonding the lens **110** to the substrate **10**, as well as optionally a hard coat, a UV blocking layer, a thermochromic film, etc. as described above.

[0042] The above description is given by way of example, and not limitation. Given the above disclosure, one skilled in the art could devise variations that are within the scope and spirit of the invention disclosed herein. Further, the various features of the embodiments disclosed herein can be used alone, or in varying combinations with each other and are not intended to be limited to the specific combination described herein. Thus, the scope of the claims is not to be limited by the illustrated embodiments.

Claims

1-30. (canceled)

31. A protective barrier affixable to a curved substrate, the protective barrier comprising a stack of one or more lenses, each of the one or more lenses including a thermoplastic polyurethane (TPU) film having a visible light transmittance (VLT) of greater than 85% and an adhesive layer on a first side of the TPU film.

32. The protective barrier of claim 31, wherein the TPU film of each of the one or more lenses has a haze of 1% or lower.

33. The protective barrier of claim 31, wherein each of the one or more lenses further includes a hard coat.

34. The protective barrier of claim 31, wherein each of the one or more lenses further includes an ultraviolet (UV) blocking layer.

35. The protective barrier of claim 31, wherein each of the one or more lenses further includes a thermochromic film.

36. The protective barrier of claim 31, wherein the TPU film of each of the one or more lenses has a thickness of greater than 2 mils.

37. A protective barrier affixable to a curved substrate, the protective barrier comprising a stack of two or more lenses, each of the two or more lenses including a thermoplastic polyurethane (TPU) film having a visible light transmittance (VLT) of greater than 85% and an adhesive layer on a first side of the TPU film.

38. The protective barrier of claim 37, wherein the TPU film of each of the two or more lenses has a haze of 1% or lower.

39. The protective barrier of claim 37, wherein each of the two or more lenses further includes a hard coat.

40. The protective barrier of claim 37, wherein each of the two or more lenses further includes an ultraviolet (UV) blocking layer.

- 41.** The protective barrier of claim 37, wherein each of the two or more lenses further includes a thermochromic film.
- 42.** The protective barrier of claim 37, wherein the TPU films of the two or more lenses have respective thicknesses that include at least two different thicknesses.
- 43.** The protective barrier of claim 42, wherein the TPU film of an innermost lens of the stack has a greater thickness than the TPU film of an outermost lens of the stack.
- 44.** The protective barrier of claim 43, wherein the TPU film of the innermost lens of the stack is at least twice as thick as the TPU film of the outermost lens of the stack.
- 45.** The protective barrier of claim 44, wherein the TPU film of the innermost lens of the stack is at least three times as thick as the TPU film of the outermost lens of the stack.
- 46.** The protective barrier of claim 37, wherein the TPU film of each of the two or more lenses has a thickness of two mil or greater.
- 47.** The protective barrier of claim 46, wherein the TPU film of an innermost lens of the stack has a thickness of seven mil or less.
- 48.** The protective barrier of claim 47, wherein the thickness of the TPU film of the innermost lens of the stack is five mil or greater.
- 49.** The protective barrier of claim 47, wherein the TPU film of an outermost lens of the stack has a thickness of three mil or less.
- 50.** A protective barrier affixable to a curved substrate, the protective barrier comprising a stack of three or more lenses, each of the three or more lenses including a thermoplastic polyurethane (TPU) film having a visible light transmittance (VLT) of greater than 85% and an adhesive layer on a first side of the TPU film.
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